16-Puzzle:

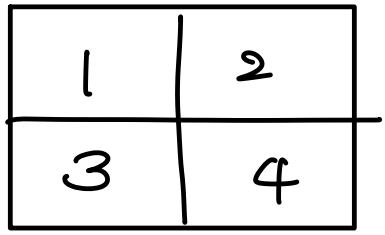
Brute-Force:

10,461,394,944,000 possibilities.

$$= \frac{16!}{2}$$

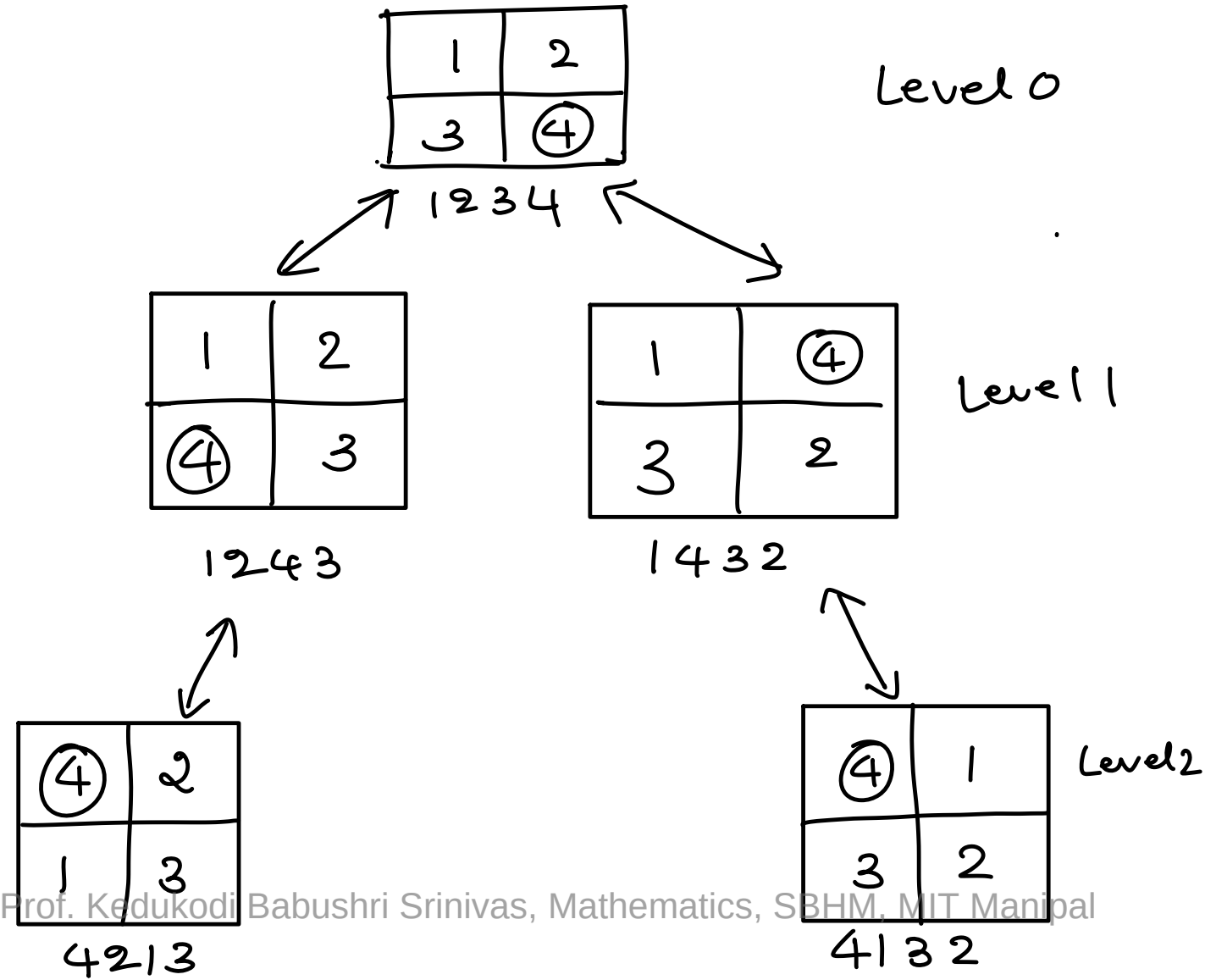
1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

4-Puzzle.



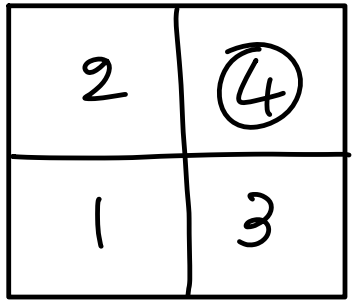
$\longleftrightarrow$ 1234

$4! = 24$ possibilities.



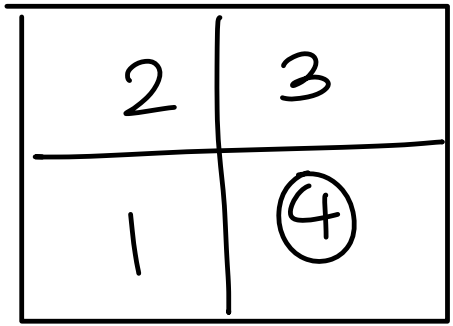
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2413



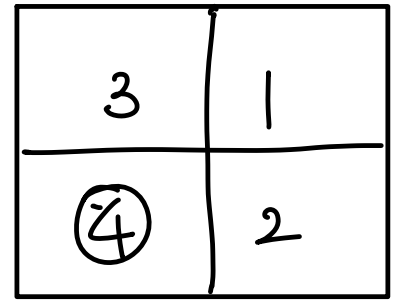


2314





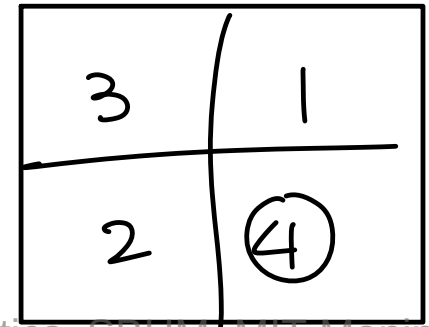
Level 3



3412



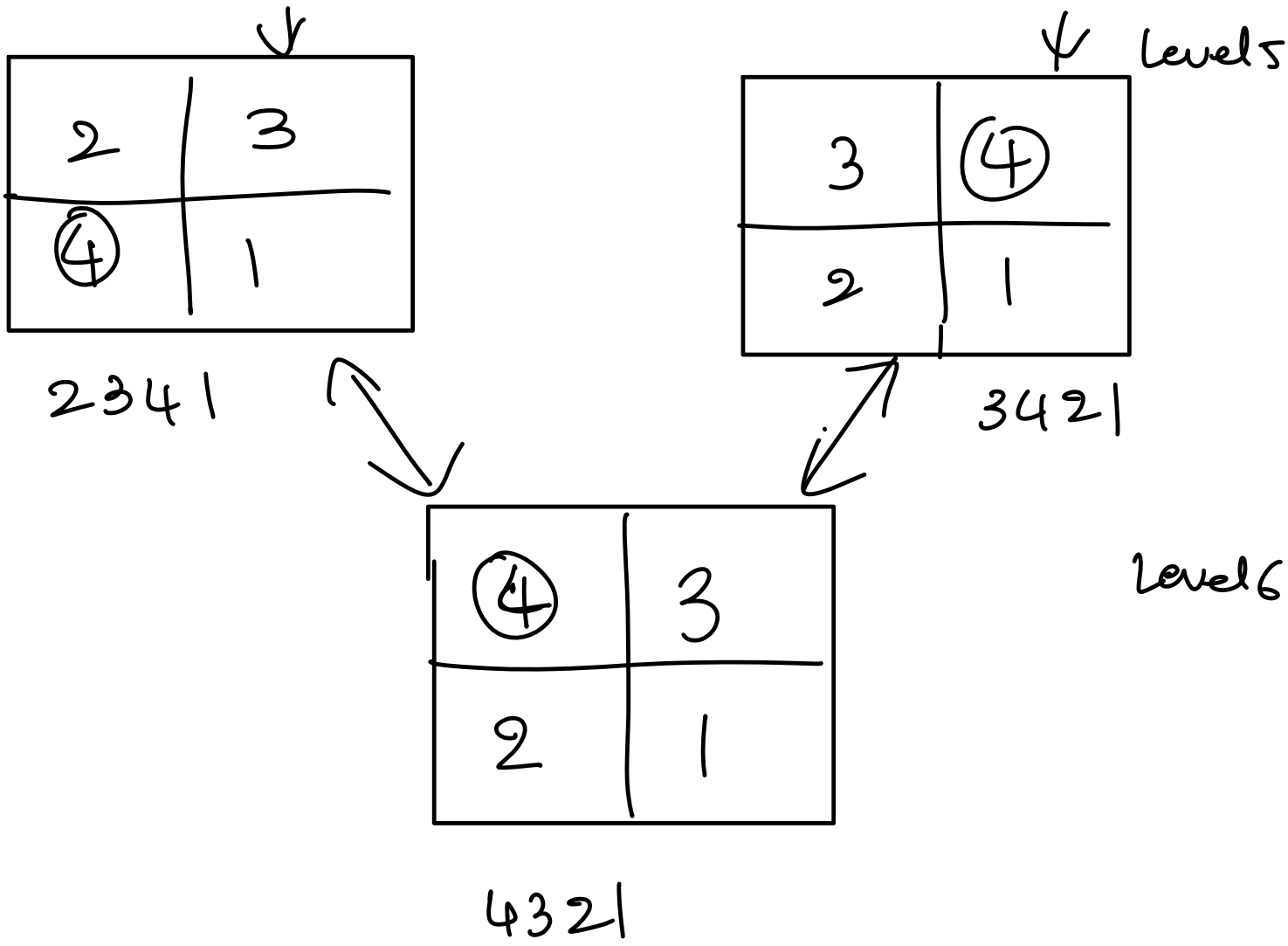
Level 4

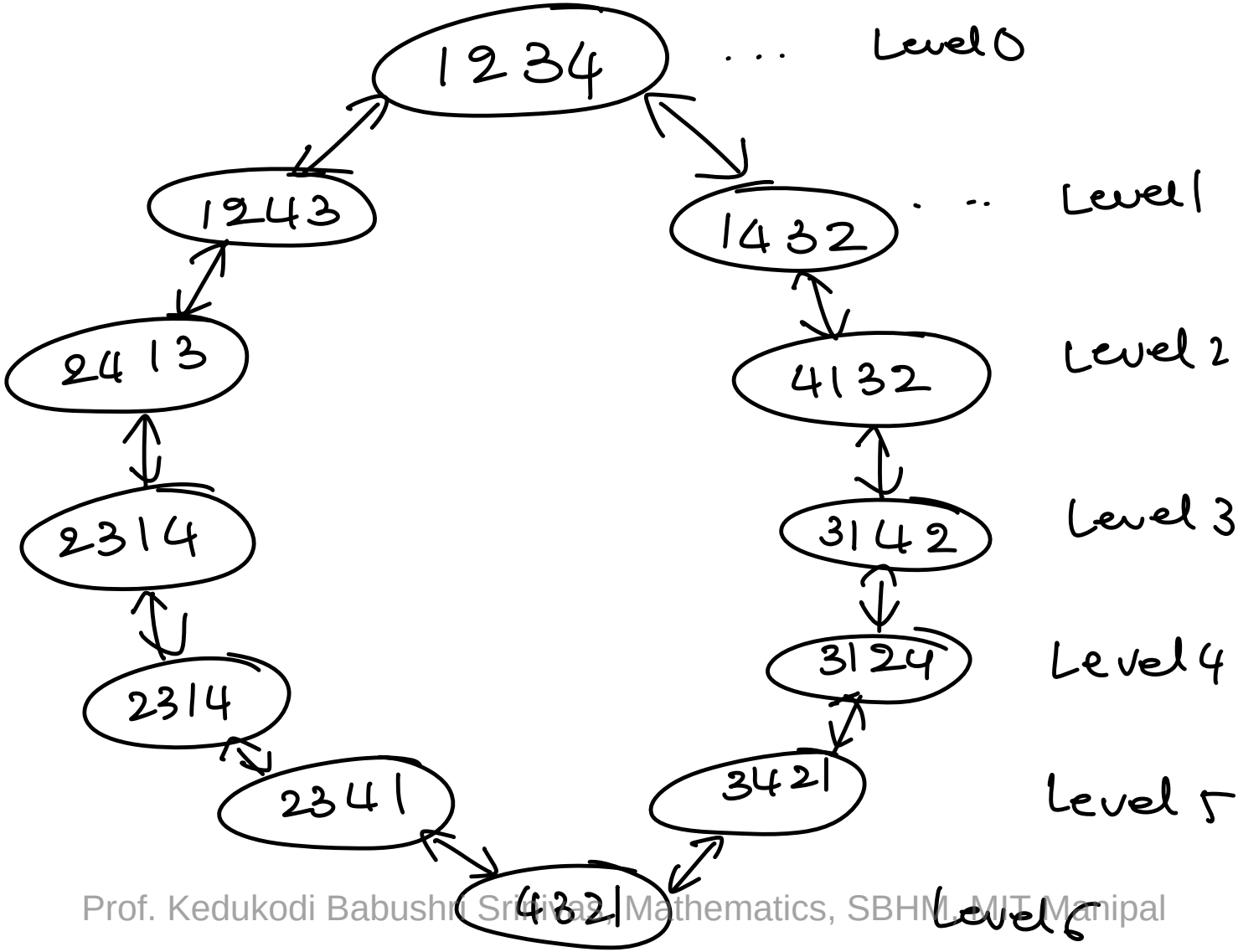


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Levels
2 3
4 1
2341
3 4
2 1
3421
4 3
2 1
4321
Level 6


1234
Level 0
1243
1432
Level 1
2413
4132
Level 2
2314
3142
Level 3
2314
3124
Level 4
2341
3421
Level 5
4321
Level 6

Only 12 states out of $4! = 24$ are attained.

$$12 = \frac{4!}{2}$$

(dictionary) lexicographic order	State (permutation)	Factorial Digits (c_1, c_2, c_3, c_4)	Total inversions $\sum c_i$	Total inversion Even/odd (Sign)	$\chi(s)$	$I(s) = \text{sign}(s) \cdot \chi(s)$
Sr. No 0	✓ 1 2 3 4	(0, 0, 0, 0)	0	Even (+1)	+1	+1
1	✓ 1 2 4 3 (2 fixed)	(0, 0, 1, 0)	1	odd (-1)	-1	+1
2	1 3 2 4	(0, 1, 0, 0)	1	odd (-1)	+1	-1
3	1 3 4 2 (3 fixed)	(0, 1, 1, 0)	2	even (+1)	-1	-1
4	1 4 2 3	(0, 2, 0, 0)	2	even (+1)	-1	-1
5	✓ 1 4 3 2 (4 fixed)	(0, 2, 1, 0)	3	odd (-1)	-1	+1
6	2 1 3 4	(1, 0, 0, 0)	1	odd (-1)	+1	-1
7	2 1 4 3	(1, 0, 1, 0)	2	even (+1)	-1	-1
8	✓ 2 3 1 4	(1, 1, 0, 0)	2	even (+1)	+1	+1

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9	2 3 4 1	(1, 1, 1, 0)	3	odd$_{(-1)}$	−1	+1
10	2 4 1 3	(1, 2, 0, 0)	3	odd$_{(-1)}$	−1	−1
11	2 4 3 1	(1, 2, 1, 0)	4	even$_{(+1)}$	−1	+1
12	3 1 2 4	(2, 0, 0, 0)	2	even$_{(+1)}$	+1	+1
13	3 1 4 2	(2, 0, 1, 0)	3	odd$_{(-1)}$	−1	+1
14	3 2 1 4	(2, 1, 0, 0)	3	odd$_{(-1)}$	+1	−1
15	3 2 4 1	(2, 1, 1, 0)	4	even$_{(+1)}$	−1	−1
16	3 4 1 2	(2, 2, 0, 0)	4	even$_{(+1)}$	−1	−1
17	3 4 2 1	(2, 2, 1, 0)	5	odd$_{(-1)}$	−1	+1
18	4 1 2 3	(3, 0, 0, 0)	3	odd$_{(-1)}$	+1	−1
19	4 1 3 2	(3, 0, 1, 0)	4	even$_{(+1)}$	+1	+1
20	4 2 1 3	(3, 1, 0, 0)	4	even$_{(+1)}$	+1	+1

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21	4 2 3 1	(3, 1, 1, 0)	5	$\text{odd}_{(-1)}$	+1	−1
22	4 3 1 2	(3, 2, 0, 0)	5	$\text{odd}_{(-1)}$	+1	−1
23	~~4~~ 3 2 1	(3, 2, 1, 0)	6	$\text{even}_{(+1)}$	+1	+1

✓ $I(S) = +1$

else $= -1$

✓ — 12 states attained by 4-puzzle

Finding Factorial Digits:

4132 (19)

$= (3)(3!) + (0)(2!) + (1)(1!) + (0)\,0!$

18 + 1 = 19

3010 is factorial digit of 4132

$4 \leq c_3 \leq 4$

$1432_{(5)}$

$= 0(3!) + (2)2! + (1)1! + 0(0!)$

$4 \quad + 1 = 5$

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Inversion method :

1432

Count of no's smaller than

1 = 0

4 = 2

3 = 1

2 = 0

Factorial Digits = (0, 2, 1, 0).

4 1 3 2

	Count of no's smaller than
4	3
1	0
3	1
2	0

(3, 0, 1, 0)

$(-1)^{1+1}$ +	$(-1)^{1+2}$ −
$(-1)^{2+1}$ −	$(-1)^{2+2}$ +

$(-1)^{\text{row + column}}$

eg:

4	1
3	2

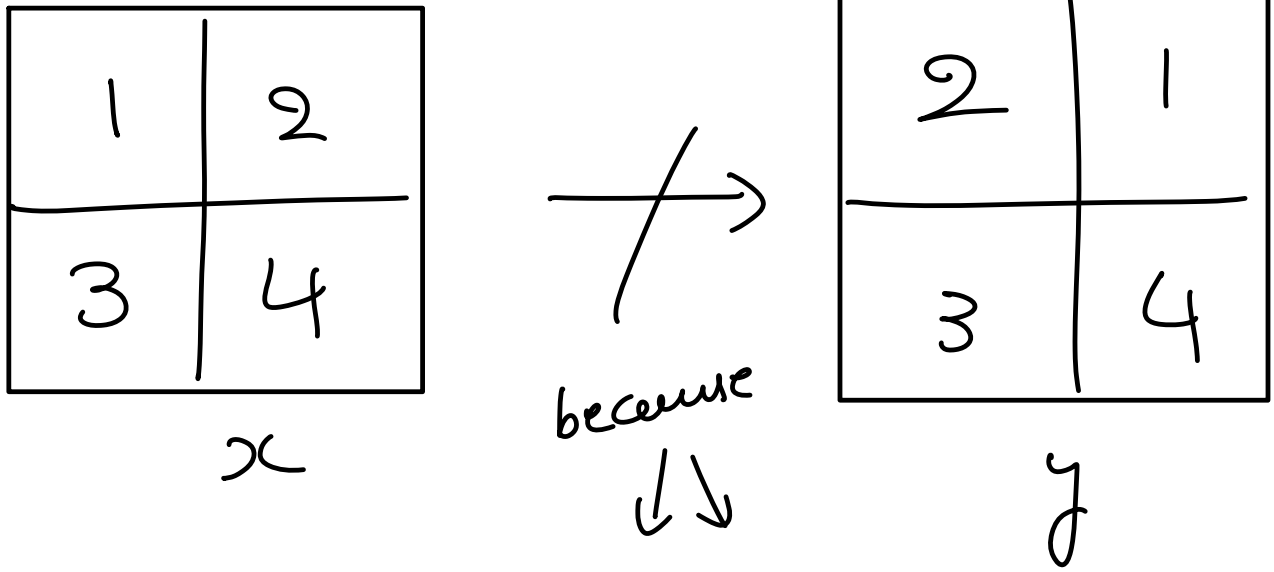
State S

$\chi(S) = +$

Sign of 4.

Def^{n}. For a state s

$$\chi(s) = \begin{cases} +1 & \text{if blank is in position 1 or 4} \\ -1 & \text{otherwise} \end{cases}$$



$$I(x) = \text{sign}(x) \cdot \chi(x)$$

$$= +1$$

$$I(y) = (-1) \cdot +1$$

$$= -1$$

16-puzzle:

x

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

?

↛

No.

$\therefore$

$I(x) = -1$

$I(y) = +1$

y

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

(Solved state)

(invariant)

$I(x)$:

$\text{sign}(x) \cdot \chi(x)$

$-1 \cdot +1 = -1$

$I(y)$:

$\text{sign}(y) \cdot \chi(y)$

$+1 \cdot +1 = 1$

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only 1 inversion (due to 15)

blank at its position 16.

No inversion